

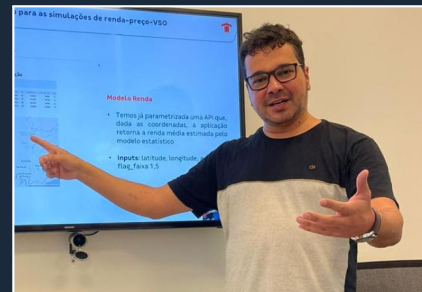
LUCAS ALVES DE SOUZA

Data Scientist | PhD in Physics

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Data Scientist with a PhD in Physics and a strong foundation in applied mathematics. I develop statistical models, machine learning pipelines, and strategic analyses that transform complex datasets into business decisions.



Data Science Experience

Data Scientist – Specialist level

9/2021 – present

Tenda, Brazil

- Data Scientist Specialist in Market Intelligence for residential real estate, supporting pricing, sales performance, regional studies, and executive decision-making.
- Contributes to machine learning-based dynamic pricing initiatives through model monitoring, analytical validation, and active participation in strategic pricing discussions.
- Develops real-time dashboards to track price components, sales volume, product mix, and market dynamics, applying advanced statistical methods to extract actionable insights.
- Builds clustering models for micro-region classification using sociodemographic, geographic, and market variables to assess regional quality and attractiveness.
- Conducts advanced price elasticity studies to quantify the relationship between price, sales velocity, demand behavior, and revenue potential.
- Creates executive-level presentations for directors and senior stakeholders, transforming complex analyses into clear recommendations for business strategy.

Research Experience

Postdoc researcher

2020 – Investigation on short-range correlations in nuclear matter of neutron stars.

ITA , Brazil

Postdoc researcher

2019 – Theoretical study on nuclear reactions of neutron-rich nuclei.

USP, Brazil

Education

2017 **Doctorate in Physics**

2011 **Master's degree in Physics**

2008 **Physics Undergraduate**

ITA , Brazil

UFSJ, Brazil

UFSJ, Brazil

Selected Training (notes/scripts → [Github](#))

2025 [Statistics for Data Science and Machine Learning](#)

2022 [TensorFlow: Machine Learning and Deep Learning with Python](#)

2022 [Genetic Algorithm in Python](#)

2021 [SQL Bootcamp](#)

2021 [Machine Learning and Data Science with Python](#)

2017 [School on few-body physics](#)

IA Expert Academy / Udemy
IA Expert Academy
IA Expert Academy
Udemy
IA Expert Academy / Udemy
MPIPKS, Dresden, Germany 

Teaching Experience (2/5 → [CV Lattes](#))

2019 **Professor of Physics and Math, 30h**

2018 **Temporary Professor of Physics, 40h**

Unitau, Brazil

UFLA, Brazil

Relevant Publications (2/14 → [Google Scholar](#))

- S. Pilling, M. T. Pazzianotto, L. A. Souza, [ApJ](#), **921**, 116(2021).
- L. A. Souza, E. V. Chimanski, B. V. Carlson, [Phys. Rev. C](#), **104**, 034623(2021).

Technical Interests

- Quantitative modeling, statistical inference, probabilistic simulation, and time series analysis.
- Machine learning, scientific computing, numerical methods, and reproducible analytical pipelines.
- AI agents, reusable skills, and LLM-assisted analytical workflows.
- Pricing analytics, market intelligence, result interpretation, and causal/temporal analysis.